

PII priority recall — 1,000-trial run

Results

The selected artifact is the **1,000-character per-label fusion**. It passed all **55/55 measurable priority tag–corpus recall gates conclusively**: every point recall was at least 0.90 and every 95% document-bootstrap lower bound was at least 0.90. Worst observed priority recall was **0.9888** and the lowest lower bound was **0.9811**.

It did not reach the quality aspiration: equal-corpus macro F2 is **0.4835** (95% corpus-bootstrap interval **0.3501–0.5892**) versus the 0.90 target. The micro F1 tie-break is **0.3812** (95% CI **0.2347–0.5348**). This is therefore a high-recall internal/research candidate, not a precision-ready production redactor.

Model / operating point	Macro F2	Micro F1	Priority point gates	Worst recall
Current rules baseline	0.0985	0.2000	2/55	0.0000
Hash recall-max, 20k	0.4234	0.3452	55/55	0.9966
Hash F2, 20k	0.4871	0.5098	49/55	0.1800
TF-IDF linear, 20k	0.4832	0.4929	50/55	0.1600
EmbeddingBag ASL, 20k	0.4647	0.4691	50/55	0.4200
Two-head priority hybrid, 20k	0.4466	0.3634	55/55	0.9966
Full per-label fusion, 20k	0.4798	0.3791	55/55	0.9966
Selected full fusion, 1k	0.4835	0.3812	55/55	0.9888

Requested read-pass speed comparison

One-core end-to-end latency was measured on the same deterministic stratified sample of 1,000 documents. Quality was re-scored on all eight holdouts at each read depth. `c_read` is the text actually read by the model.

Read pass	p50 ms	p95 ms	p99 ms	Mean ms	Docs /s	p95 speed vs 20k	p95 saved vs 20k	Macro F2	Micro F1	Priority gates	Worst recall
1,000	0.825	2.200	2.386	1.088	919.1	4.01×	6.629 ms	0.4835	0.3812	55/55	0.9888
2,500	1.055	4.211	4.620	1.754	570.1	2.10×	4.618 ms	0.4797	0.3801	55/55	0.9686
10,000	1.075	6.766	8.220	2.440	409.9	1.30×	2.063 ms	0.4799	0.3792	55/55	0.9966
20,000	1.064	8.829	11.885	2.822	354.4	1.00×	0.000 ms	0.4798	0.3791	55/55	0.9966

The 1k pass is **4.01× faster at p95** than 20k and saves 6.63 ms/document. It also has slightly higher macro F2 (+0.0036), but lowers the worst priority recall by 0.0079. The 1k artifact passes the fast gate because its effective read ceiling is 1k and p95 is 2.20 ms. Full 10k and 20k fusion passes exceed the 5 ms p95 aspiration (6.77 and 8.83 ms); they must not be described as meeting it.

Holdout matrix, one corpus at a time

Holdout corpus	Rows	Gold mode	Macro F2	Micro F1	Conclusive priority gates	Worst recall
4000_datax-dualjudge-evalset-1.32k	4,000	Positive-only	N/A	N/A	1/1	1.0000
6589_govdocs2-dualjudge-eval20-3.53k	6,589	Positive-only	N/A	N/A	2/2	0.9888
ai4privacy_pii_masking_eval_10k	10,626	Complete	0.4995	0.3529	10/10	0.9991
betterdataai_ner_silver_eval_10k	10,360	Complete	0.2313	0.2022	4/4	1.0000
nemotron_eval_6k	5,617	Positive-only	N/A	N/A	0/0	N/A
openpii_pii_eval_38k	38,937	Complete	0.6297	0.6613	9/9	0.9997
pii2_eval_30k	30,000	Complete	0.4930	0.2082	15/15	0.9958
pii_holdout_20k	20,000	Complete	0.5638	0.4813	14/14	0.9944

OpenPII is the most flattering complete corpus and BetterDataAI is the weakest. The number to quote across corpora is the equal-corpus macro F2 of 0.4835, not OpenPII's 0.6297.

Priority-tag evidence

Each row is the minimum across separately scored corpora having at least 30 positives for that tag. The confidence value is the lowest lower 95% bootstrap bound across those corpora.

Priority tag	Corpora measured	Total support	Minimum recall	Lowest 95% bound	Result
Social Security number	4	12,551	0.9958	0.9861	PASS
ITIN	3	10,543	1.0000	1.0000	PASS
Medical record number	3	1,104	1.0000	1.0000	PASS

Priority tag	Corpora measured	Total support	Minimum recall	Lowest 95% bound	Result
Health-plan beneficiary number	2	890	0.9944	0.9850	PASS
Patient ID	1	326	1.0000	1.0000	PASS
Bank account number	3	1,438	0.9983	0.9946	PASS
Credit-card number	4	12,156	1.0000	1.0000	PASS
IBAN	2	740	1.0000	1.0000	PASS
Passport number	4	19,419	0.9991	0.9971	PASS
Driver's licence number	4	12,666	1.0000	1.0000	PASS
Military ID	2	18,090	0.9998	0.9996	PASS
Visa number	3	18,317	0.9998	0.9995	PASS
Password	4	1,723	0.9992	0.9975	PASS
PIN	3	613	1.0000	1.0000	PASS
Full name	6	60,902	0.9972	0.9952	PASS
Address	7	26,473	0.9888	0.9811	PASS

TL;DR

- Exactly **1,000** MLflow-tracked trials were completed: 300 hash, 300 TF-IDF, 300 EmbeddingBag ASL, and 100 fusion candidates.
- The selected 1k fusion passed every measurable 90% priority-recall gate with conclusive 95% bootstrap evidence and runs at 2.20 ms p95 on one core.
- It is 4.01× faster at p95 than the same 20k fusion.
- Macro F2 remains only 0.4835, so use it for high-recall triage with downstream review—not as a stand-alone redactor or proof that a document is PII-free.

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
current_rules	4000_datax-dualjudge-evalset-1.32k					
current_rules	6589_govdocs2-dualjudge-eval20-3.53k					
current_rules	ai4privacy_pii_masking_eval_10k					
current_rules	betterdataai_ner_silver_eval_10k					
current_rules	nemotron_eval_6k					
current_rules	openpii_pii_eval_38k					
current_rules	pii2_eval_30k					
current_rules	pii_holdout_20k					
hash_sgd	4000_datax-dualjudge-evalset-1.32k					
hash_sgd	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd	ai4privacy_pii_masking_eval_10k					
hash_sgd	betterdataai_ner_silver_eval_10k					
hash_sgd	nemotron_eval_6k					
hash_sgd	openpii_pii_eval_38k					
hash_sgd	pii2_eval_30k					
hash_sgd	pii_holdout_20k					
hash_sgd_f2	4000_datax-dualjudge-evalset-1.32k					
hash_sgd_f2	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd_f2	ai4privacy_pii_masking_eval_10k					
hash_sgd_f2	betterdataai_ner_silver_eval_10k					
hash_sgd_f2	nemotron_eval_6k					
hash_sgd_f2	openpii_pii_eval_38k					
hash_sgd_f2	pii2_eval_30k					
hash_sgd_f2	pii_holdout_20k					
hybrid_priority_001	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority_001	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority_001	ai4privacy_pii_masking_eval_10k					
hybrid_priority_001	betterdataai_ner_silver_eval_10k					

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
hybrid_priority_001	nemotron_eval_6k					
hybrid_priority_001	openpii_pii_eval_38k					
hybrid_priority_001	pii2_eval_30k					
hybrid_priority_001	pii_holdout_20k					
tfidf_linear	4000_datax-dualjudge-evalset-1.32k					
tfidf_linear	6589_govdocs2-dualjudge-eval20-3.53k					
tfidf_linear	ai4privacy_pii_masking_eval_10k					
tfidf_linear	betterdataai_ner_silver_eval_10k					
tfidf_linear	nemotron_eval_6k					
tfidf_linear	openpii_pii_eval_38k					
tfidf_linear	pii2_eval_30k					
tfidf_linear	pii_holdout_20k					
embeddingbag_asl	4000_datax-dualjudge-evalset-1.32k					
embeddingbag_asl	6589_govdocs2-dualjudge-eval20-3.53k					
embeddingbag_asl	ai4privacy_pii_masking_eval_10k					
embeddingbag_asl	betterdataai_ner_silver_eval_10k					
embeddingbag_asl	nemotron_eval_6k					
embeddingbag_asl	openpii_pii_eval_38k					
embeddingbag_asl	pii2_eval_30k					
embeddingbag_asl	pii_holdout_20k					
hybrid_priority	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority	ai4privacy_pii_masking_eval_10k					
hybrid_priority	betterdataai_ner_silver_eval_10k					
hybrid_priority	nemotron_eval_6k					
hybrid_priority	openpii_pii_eval_38k					
hybrid_priority	pii2_eval_30k					
hybrid_priority	pii_holdout_20k					
champion_1k	4000_datax-dualjudge-evalset-1.32k					

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Docs/s ec	ms/ doc
champion_1k	6589_govdocs2-dualjudge-eval20-3.53k					
champion_1k	ai4privacy_pii_masking_eval_10k					
champion_1k	betterdataai_ner_silver_eval_10k					
champion_1k	nemotron_eval_6k					
champion_1k	openpii_pii_eval_38k					
champion_1k	pii2_eval_30k					
champion_1k	pii_holdout_20k					
champion_1k_bootstrap	8 holdouts (5 complete for F2)				454.5	2.2
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				984.8	1.0
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				483.5	2.1
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				204.6	4.9
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				146.2	6.8
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				674.0	1.5
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				341.6	2.9
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				175.4	5.7
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				130.7	7.6
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				454.5	2.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				237.5	4.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				147.8	6.8
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				113.3	8.8

Measured 2026-08-25 · n=1,000, n=126,129.